

# CECOS University

## PPIT Project

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**Course:** Professional Practices in IT

**Institution:** CECOS University Peshawar

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# 1 Lecture 1: Introduction to Professional Issues in IT

## 1. Why does CECOS University consider Professional Issues in IT essential?

**Answer:**

CECOS believes that technical knowledge alone is not enough to create responsible computing professionals. The course develops understanding of ethics, law, communication, and professional conduct, ensuring graduates act with integrity and accountability.

## 2. How does the combination of laws, ethics, and management topics help students?

**Answer:**

Students learn to apply ethical principles, understand legal implications, and use management skills effectively. This integration shapes graduates who can lead responsibly in technical and managerial roles.

## 3. What are the real-world consequences if such a course were not included?

**Answer:**

Without this foundation, graduates might violate cyber laws, misuse data, or engage in unethical practices, leading to legal penalties, job loss, or loss of credibility.

#### **4. How does CECOS ensure ethical understanding when developing new technologies?**

**Answer:**

Through discussions, seminars, and case studies on AI ethics, data bias, and privacy concerns, encouraging students to evaluate the social impact of emerging technologies.

#### **5. How do subjects like Professional Ethics, Technical Writing, and Pakistan Studies help?**

**Answer:**

They build moral awareness, communication proficiency, and national understanding, producing graduates who act ethically and contribute positively to society.

#### **6. How does this course improve employability and readiness?**

**Answer:**

Employers value graduates who understand professionalism, law, and teamwork. The course ensures students are industry-ready and trustworthy in professional environments.

## 2 Lecture 2: Laws & Legal Systems

### 1. Which IT-related laws are included?

**Answer:**

Electronic Transactions Ordinance 2002, PECA 2016, FIA Cybercrime Regulations, Data Protection, and Intellectual Property Laws.

### 2. Does CECOS conduct seminars on cyber laws?

**Answer:**

Yes, in collaboration with FIA Cyber Crime Wing and NADRA experts, to educate students about digital laws and privacy.

### 3. How does CECOS explain civil vs. criminal laws?

**Answer:**

Through real examples: civil (copyright, defamation) vs. criminal (hacking, identity theft). Mock trials and case studies are used.

### 4. What real cybercrime cases are discussed?

**Answer:**

Bank hacking, social media impersonation, and data leaks from e-commerce platforms are analyzed as case studies.

### 5. What role does FIA play?

**Answer:**

FIA collaborates through awareness sessions, internships, and investigation briefings.

## **6. Has any CECOS student worked on cybercrime prevention?**

### **Answer:**

Yes, through Final Year Projects on intrusion detection, phishing prevention, and data encryption.

## **7. How does CECOS balance privacy and security in ethics?**

### **Answer:**

Students discuss both perspectives to reach balanced decisions guided by ethical frameworks.

## 3 Lecture 3: Professionalism, Accreditation & Ethics

### 1. Difference between professional and ordinary employee?

**Answer:**

A professional combines technical skill with ethical judgment; an ordinary employee only performs tasks.

### 2. How does CECOS ensure ethical and professional standards?

**Answer:**

Via mandatory courses, plagiarism checks, mentorship, and code-of-conduct policies aligned with IEEE/ACM.

### 3. Is CECOS accredited by NCEAC or PEC?

**Answer:**

Yes, computing programs by NCEAC and engineering programs by PEC, ensuring global degree recognition.

### 4. What are the policies for plagiarism?

**Answer:**

Zero tolerance: offenders face grade penalties, warnings, or expulsion.

## **5. Does CECOS have IEEE or ACM chapters?**

**Answer:**

Yes, IEEE Student Chapter organizes workshops, competitions, and conferences to build global exposure.

## **6. How does CECOS prepare students for international certifications?**

**Answer:**

By guiding students toward certifications (AWS, Cisco, Microsoft) and memberships (IEEE, ACM, ISACA).

## **7. Does Pakistan need a stronger code of ethics?**

**Answer:**

Yes, to standardize conduct across IT professions similar to medical or legal codes.



## 4 Lecture 4: Organization, Registration & Legal Framework

### 1. What is the legal status of CECOS University?

**Answer:**

A chartered private university established under the CECOS University Act 2001 by the Government of Khyber Pakhtunkhwa.

### 2. Under which authority is CECOS recognized?

**Answer:**

Recognized by HEC, Government of Khyber Pakhtunkhwa, and registered with SECP.

### 3. Who are the trustees or board members?

**Answer:**

Managed by the CECOS Educational Institutions (CEI) Trust, led by Engr. Muhammad Tanveer Javed, along with a Board of Trustees and Academic Council.

### 4. How does CECOS ensure compliance with regulations?

**Answer:**

Through annual audits, academic reviews, and compliance reports submitted to SECP and HEC.

## **5. Is CECOS registered with PSEB?**

**Answer:**

Collaborates with PSEB for training and software projects, though primarily an academic institution.

## **6. How is the university funded?**

**Answer:**

Through tuition fees, private donors, and research grants; transparency ensured via independent audits.

## **7. Does CECOS collaborate with industry or international partners?**

**Answer:**

Yes, with Microsoft, Huawei, and local firms for internships, plus partnerships with global universities for research and exchange programs.

## 5 Chapter 5: Organizational Governance

### 1. What governance structure does CECOS follow — single-tier or two-tier?

**Answer:**

CECOS follows a two-tier structure. The university's governance separates supervisory oversight from operational management: a Board of Governors provides high-level supervision and strategic oversight, while an Executive Management team is responsible for day-to-day operations and implementation of policy. This split allows for checks and balances between governance and management functions and supports clear accountability.

### 2. How does the university manage conflict of interest situations?

**Answer:**

CECOS manages conflict of interest situations through formal policies and procedures. These include a Conduct and Discipline Policy and an Anti-Harassment Policy, plus specific protections and procedures such as Protection against Harassment of Women at Workplace and harassment inquiry procedures. Compliance is further reinforced by the University Charter of Social Responsibility, which sets institutional expectations for ethical conduct and disclosures. In practice, suspected conflicts are addressed through established inquiry and investigation steps defined in these policies so that due process and impartiality are maintained.

### 3. What is the key difference between executive and non-executive roles here?

**Answer:**

*Executive roles* are held by full-time employees who manage operational activities — for example the Vice Chancellor, Deans, and Directors who run academic and administrative units. *Non-executive roles* are typically external stakeholders who provide oversight without day-to-day operational responsibilities, such as board members from external institutions or HEC representatives. To make this distinction clearer, a comparison table is provided below.

**Table 5.1: Executive vs. Non-Executive Roles**

| <b>Executive Roles (Operational)</b>                                                   | <b>Non-Executive Roles (Oversight)</b>                                              |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Vice Chancellor, Deans, Directors: full-time management responsible for implementation | Board members, HEC reps: external oversight, policy review, and governance guidance |
| Involved in daily decision-making and resource allocation                              | Provide strategic oversight and accountability; do not engage in daily operations   |
| Accountable for operational performance                                                | Accountable for governance and compliance                                           |

### 4. Which organizational structure does CECOS use (hierarchical, matrix, or hybrid)?

**Answer:**

CECOS uses a hierarchical structure organized largely by function. The chain of command is typically: Board → Vice Chancellor → Department Heads → Staff. Academic activities are grouped into departments (twelve academic departments), and administrative centers support university functions. This structure provides clarity in reporting lines and responsibility.

## 5. Who decides resource allocation for academic and administrative projects?

### **Answer:**

Resource allocation is primarily a centralized function. The Vice Chancellor acts as the primary authority for routine allocations, while major budgetary decisions and strategic funding choices are approved by the Board of Governors. Administrative resources are coordinated by the Vice President Administration & Coordination. In practice, proposals from departments are reviewed upward through this chain for approval and funding.

## 6. Is decision-making centralized or decentralized across departments?

### **Answer:**

Decision-making at CECOS is primarily centralized. Strategic and policy decisions are made at the top levels of the university, and the Quality Enhancement Cell (QEC) ensures consistent university-wide standards. Departments have limited autonomy within the budgets and policies allocated to them; operational flexibility exists but within the centrally determined rules for admission, examinations, and discipline.

## 7. How is accountability handled when a department faces an issue or failure?

### **Answer:**

Accountability follows a clear chain: Department Head → Dean/Supervisor → Vice Chancellor → Board of Governors. The Quality Enhancement Cell monitors compliance and quality indicators; formal investigations are carried out under the Conduct and Discipline Policy when needed. External accountability is provided by regulatory bodies such as HEC where applicable, and the Board reports to stakeholders about corrective actions and outcomes.

## 8. How are hiring and promotions decided here?

### **Answer:**

**Hiring:** Hiring follows merit-based selection procedures. Candidates undergo formal interviews and screening; senior appointments require Vice Chancellor approval and top-level positions require Board approval. The process is designed to ensure fairness and transparency.

**Promotions:** Promotions are based on seniority and achievement, guided by the Faculty Retention Policy 2023. Annual performance evaluations and advanced qualifications (MS/PhD) are considered in promotion decisions. A formal review process is used to minimize arbitrary decisions and to protect faculty rights.

## **6 Chapter 6: Funding & Entrepreneurship**

### **1. Does CECOS help students prepare a proper business plan for funding?**

**Answer:**

CECOS teaches students how to structure and write a business plan. Students learn market analysis, financial planning, and idea development through coursework and project work. Guidance is provided by faculty and through supervised projects, and students prepare their own plans applying the concepts taught in class.

### **2. What funding options do you recommend for new startups — grants, loans, or equity?**

**Answer:**

Grants are recommended as the safest initial option since they do not require repayment. Government soft loans are suitable for small tech startups emerging from the university. Equity investment is useful for larger, scalable ideas that require significant capital; however, it involves sharing ownership. The general advice is to start with grants or soft loans, and move to equity only when scaling demands larger funds.

### **3. Does the university assist students in finding or applying for grants?**

**Answer:**

CECOS provides awareness about available grants and funding programs through seminars and faculty guidance. Teachers introduce students to national grant schemes and funding opportunities, but the students are expected to search and apply for grants themselves. The university assists by offering information and mentorship rather than

applying on behalf of students.

#### **4. How does CECOS guide students about bank loans, overdrafts, or soft loans?**

**Answer:**

Students are taught the workings of bank loans, overdrafts, and soft loans as part of entrepreneurship-related material. Instruction covers interest calculations, repayment schedules, security requirements, and suitability of different loan types for business needs. This financial literacy prepares students to approach banks confidently and choose appropriate financing options.

#### **5. Do you teach students how to raise equity from investors or business angels?**

**Answer:**

Yes. CECOS includes the concept of equity financing in entrepreneurship courses. Students learn how investors provide funds in exchange for ownership, and events/exhibitions expose students to industry professionals. The focus is primarily theoretical and preparatory: students gain understanding of investor expectations and the process of raising equity.

#### **6. How should a new entrepreneur decide between taking a loan or selling equity?**

**Answer:**

The decision depends on the entrepreneur's goals. Loans are recommended if the founder wants to retain full control of the company; equity is appropriate for ventures that require large-scale investment and that the founders are willing to share ownership. The choice should consider cash flow projections, repayment capacity, long-term growth goals, and the impact of ownership dilution.



## 7. How do you explain the risk of high leverage (more loans) to students?

### Answer:

High leverage increases financial pressure because a larger share of profits must be used for interest payments. Lenders may be reluctant to support startups with excessive debt. Excessive borrowing reduces a company's stability and survival prospects. CECOS therefore emphasizes responsible financial planning to avoid overloading startups with debt.

**Table 6.1: Comparison — Loans vs Equity**

| Loans                                                      | Equity                                                               |
|------------------------------------------------------------|----------------------------------------------------------------------|
| Retain full control of ownership; must repay with interest | Give up partial ownership; no repayment obligation but share profits |
| Suitable for businesses with predictable cash flow         | Suitable for high-growth scalable startups needing large capital     |
| Risk of insolvency if cash flows do not cover payments     | Risk of dilution of control and decision-making power                |

## 7 Chapter 7: Ethics in IT

### 1. Which ethical issues in IT does CECOS consider most important (privacy, security, IP)?

**Answer:**

CECOS highlights three primary ethical concerns in IT: privacy (including data protection, algorithmic bias, and social impact), security (preventing cybercrime while balancing privacy), and intellectual property (preventing plagiarism, unauthorized copying of software, and protecting innovation integrity). The following table summarizes these focus areas.

**Table 7.1: Key Ethical Issues in IT**

| Privacy                                                    | Security                                                 | IP                                                               |
|------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------|
| Data protection, consent, handling of personal information | Cybercrime prevention, secure systems, incident response | Plagiarism prevention, software licensing, protecting innovation |

### 2. How does CECOS train students to avoid or prevent cybercrime?

**Answer:**

Training is multi-layered. CECOS runs seminars with organizations like FIA and NADRA on PECA 2016 and cyber laws; uses case studies (bank hacks, phishing, data leaks) as teaching material; supervises FYPs focused on intrusion detection systems, encryption, and phishing prevention; and conducts mock trials and exercises to expose students to real-world scenarios and ethical decision-making.

### **3. What policies does CECOS follow regarding piracy, plagiarism, and software copying?**

**Answer:**

CECOS follows a zero-tolerance policy on academic and software piracy. Assignments and theses are checked with Turnitin and similar tools; penalties for confirmed plagiarism can include grade penalties up to expulsion. Software piracy and unauthorized copying are prohibited under institutional code-of-conduct and relevant IP laws. Enforcement aligns with professional codes such as IEEE and ACM.

### **4. How does CECOS address ethical concerns about AI decision-making?**

**Answer:**

CECOS addresses AI ethics through curriculum content and seminar discussions. Topics include algorithmic bias, transparency and explainability, accountability, and privacy concerns in AI systems. Case studies are used to evaluate the social impact of AI, and FYPs may include bias-detection and mitigation components to instill practical understanding.

### **5. Why should IT graduates follow business ethics in workplace environments?**

**Answer:**

Adhering to business ethics prevents legal risks and job loss, builds credibility and trust with employers and clients, and promotes sustainable, responsible leadership. Ethical behavior aligns graduates with professional readiness goals and enhances employability and reputation.

### **6. How do you clearly differentiate between a professional gift and a bribe?**

**Answer:**

CECOS emphasizes clear distinctions:

**Table 7.2: Gift vs Bribe**

| <b>Professional Gift</b>                                         | <b>Bribe</b>                                                           |
|------------------------------------------------------------------|------------------------------------------------------------------------|
| Given as goodwill, with no expectation of influencing a decision | Given with the intent to influence a decision or gain unfair advantage |
| Modest and customary; transparent and disclosed                  | Excessive or lavish; often secretive                                   |
| Usually presented after an event or relationship is established  | Often presented before a decision or awarding of benefit               |

## **7. What process does CECOS teach for making ethical decisions in difficult situations?**

**Answer:**

CECOS teaches a 6-step ethical framework:

1. Identify the dilemma and all stakeholders involved.
2. Gather facts, including legal and contextual information.
3. Evaluate options using ethical lenses (utilitarian, duty-based, virtue ethics).
4. Consult mentors, codes of conduct (IEEE, ACM) and relevant policies.
5. Decide and act, ensuring the action is documented.
6. Reflect and review the outcome to learn and improve future decisions.

## 8 Chapter 8: Moral & Ethical Decision Making

### 1. How does CECOS teach students to make ethical decisions in real IT situations?

**Answer:**

CECOS teaches students to analyze the intensity of an ethical issue, identify relevant stakeholders, evaluate consequences, and recognize how personal values and organizational expectations influence decisions. Instruction combines theoretical models with practical exercises: students work through real IT case studies and discussions and follow a concise 4-step decision-making model to structure responses to ethical problems.

### 2. What role do CECOS teachers and management play in promoting ethical behavior?

**Answer:**

Teachers act as ethical leaders by modeling professional conduct and guiding students toward honesty, fairness, and responsibility. Management supports this by enforcing policies and ensuring institutional frameworks are in place so that professional standards are maintained across the university.

### 3. How does CECOS help students deal with ethical dilemmas in projects or internships?

**Answer:**

CECOS helps students by discussing workplace pressures and sharing case studies from industry and past internships. Faculty advise on best practices for ethical communication with supervisors and encourage students to document dilemmas and the steps taken to resolve them.

#### **4. Does CECOS teach moral theories like utilitarianism or duty-based ethics?**

**Answer:**

Yes. CECOS includes moral theories such as utilitarianism, Kantian duty-based ethics, virtue ethics, and the common-good approach in its teaching so students can apply differing ethical perspectives to practical problems.

#### **5. What moral frameworks does CECOS teach to help students decide what is right or wrong?**

**Answer:**

CECOS covers a range of frameworks: utilitarianism, egoism and enlightened egoism, Kantian ethics, virtue ethics, the common-good approach, fairness/relativist approaches, and Ghazalian ethics (with emphasis on moral development through reasoning and virtues). These frameworks equip students with multiple lenses for evaluating complex issues.

#### **6. What topics related to cyber ethics does CECOS focus on the most?**

**Answer:**

Primary cyber-ethics topics include privacy and security, cybercrimes and hacking tools, social media ethical risks, and issues of copyright and plagiarism. These areas are emphasized through coursework, seminars, and project work.

#### **7. What steps does CECOS take to prevent academic misconduct and unethical IT practices?**

**Answer:**

CECOS uses plagiarism-detection tools (e.g., Turnitin) for assignments and FYPs, enforces strict examination hall rules, applies penalties for cheating and copying, and runs awareness sessions on academic integrity to promote ethical conduct among students.

## **8. How does CECOS teach students to consider all stakeholders before making decisions?**

### **Answer:**

Students are taught to identify primary and secondary stakeholders, analyze potential impacts on society, organizations, users, and the environment, and use stakeholder-mapping tools (tables, weighted scoring) to evaluate trade-offs. Project work requires explicit stakeholder analysis and documentation.

## **9. What systems does CECOS use to prevent cheating, plagiarism, and unethical IT practices?**

### **Answer:**

Systems include plagiarism-checking software for academic submissions, proctored exams and supervised computer labs, formal monitoring and reporting mechanisms for misconduct, and an established disciplinary framework that can impose warnings, grade reductions, or disciplinary action as appropriate.